Yuli Liu

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Research Interests

My research centers on learning and decision-making for resource-constrained wireless networks, with current interests in (i) semantic and task-oriented communication and resource allocation, (ii) distributed and online learning under imperfect feedback, and (iii) coordination in UAV, maritime, and heterogeneous networked systems.

Education

University of Macau Aug. 2024 – Jun. 2026
M.Sc. in Smart Ocean Technology *Macau SAR, China*

- *Medium of Instruction: English.*
- Master's Thesis: *Topology Optimization for UAV-Relay-Enabled Underwater Acoustic Sensor Networks*; Advisor: Prof. Joel Oliveira Reis.
- Selected Coursework: Machine Learning and Applications in Ocean Science; Marine Robotics and Application; Ocean Acoustics; Cloud Computing.

China Jiliang University College of Modern Science and Technology Sep. 2020 – Jun. 2024
B.Eng. in Communication Engineering *Yiwu, China*

- Undergraduate Thesis: *Design of a Wearable Human Fall Detection System*; Advisor: Dr. Huixin Zhao.
- Selected Coursework: Signals and Systems; Principles of Communication; Digital Mobile Communications; C/C++ Programming; Embedded DSP Systems.

Research Experience

Shenzhen Research Institute of Big Data (SRIBD)- CUHK (Shenzhen) Aug. 2026 – Sep. 2026
Research Assistant, Center for Engineering Computing and Optimization *Shenzhen, China*
Project: Learning Dynamics of Flow Matching

- Study when global Gaussian structure and higher-order or rare-mode structure become visible along the flow path, and how quickly each is learned during optimization.
- Derive results using Gaussian projection, OU/Hermite filtering, posterior mode revelation, and local-kernel dynamics, and test the predictions with controlled conditional flow-matching experiments.

Shenzhen Institute of Artificial Intelligence and Robotics for Society Jul. 2025 – Jul. 2026
Research Assistant, The Chinese University of Hong Kong, Shenzhen *Shenzhen, China*

- Formulated a dependency-aware semantic distortion model for open-vocabulary scene graphs, where triplet utility depends on the reliability of prerequisite vocabulary packets.
- Jointly optimized MCS, transmit power, and HARQ depth under latency and energy constraints, and evaluated the resulting policy on wireless visual-uplink traces.
- *Related first-author manuscript:* "Dependency-Aware HARQ and Link Adaptation for Wireless Transmission of Open-Vocabulary Scene Graphs" (arXiv; submitted to **IEEE TWC**).

CAUSAL Lab, Shanghai University of Finance and Economics Sep. 2025 – Jul. 2026
Visiting Student; ongoing collaboration with Prof. Zhiheng Zhang *Shanghai, China*

- Investigated online causal decision-making in networked systems with interference, temporal dependence, and limited counterfactual support for long-run policy evaluation.
- Developed a doubly robust steady-state learning framework that integrates causal value estimation with uncertainty-aware exploration and adaptive network configuration. Studied how evolving observations can support long-term decisions while mitigating sensitivity to model misspecification, particularly under uncertain and evolving network conditions. The resulting first-author work was accepted at **NeurIPS 2026**.
- Investigated reliable causal decision-making under imperfect data and limited feedback, focusing on how estimation uncertainty and predictive errors affect robust policy selection in complex systems.

Publications and Manuscripts

Peer-Reviewed Publications

- **Liu, Y.**, & Zhang, Z. (2026). *Online Causal Configuration for Networked Systems via Doubly Robust Steady-State Learning*. **NeurIPS 2026 (Accepted)**.
- **Liu, Y.**, Dai, M., Dou, C., Li, P., Wu, Y., & Qian, L. (2025). *Low-Altitude Task Offloading for UAV–USV Collaboration in Marine Edge-Enabled Networks*. In **AIoTC 2025**, 355–361. **Best Paper Award**.

- Chen, H., Huo, S., Muddassir, M., Lee, H.-Y., **Liu, Y.**, Li, J., Duan, A., Zheng, P., & Navarro-Alarcon, D. (2025). *PSO-Based Optimal Coverage Path Planning for Surface Defect Inspection of 3C Components With a Robotic Line Scanner*. **IEEE Transactions on Instrumentation and Measurement**, 74, 1–12.

Journal Manuscripts and Preprints

- **Liu, Y.**, Ruan, J., & Sun, C. *Dependency-Aware HARQ and Link Adaptation for Wireless Transmission of Open-Vocabulary Scene Graphs*. arXiv:2608.08110. **Submitted to IEEE TWC**.
- **Liu, Y.**, & Wu, D. *Cost-Efficient End-to-End Key Delivery in Heterogeneous Space–Air–Ground–Sea QKD Networks*. **Submitted to IEEE TCOM**.
- **Liu, Y.**, & Zhang, X. *Federated PPO for Cooperative Sensing Control in Maritime UAV–USV Networks*. **Submitted to IEEE TVT**.
- **Liu, Y.**, & Zhang, X. *Hierarchical Federated Fusion: A Robust Cooperative Sensing Framework for Maritime IoT*. **Under review at IEEE IoTJ**.
- **Liu, Y.**, & Zhang, X. *Multi-Node Cooperative Sensing in Maritime Environments: Challenges and Frontier Technologies*. **Under review at IEEE Sensors Journal**.

Conference Manuscripts Under Review

Four research manuscripts currently under review at the International Conference on Learning Representations (ICLR 2027), covering:

- Reliable causal decision-making under limited evidence.
- Evidence-aware decision-making over large candidate spaces.
- Resource-efficient learning with delayed feedback.
- Online optimization in dynamic networked systems.

Collaborative Research

Reliable Online Decisions and Heterogeneous Network Optimization 2025 – Present
Collaboration with Prof. Di Wu, La Trobe University *Remote / Australia*

- Investigated reliable decision-making over large candidate spaces under limited evidence, focusing on how predictive information, estimation uncertainty, and causal support affect action selection.
- Explored resource-constrained adaptation in nonstationary network and sensor streams, studying how distribution shifts, delayed feedback, and limited model updates affect predictive performance.
- Investigated online optimization with evolving feasible actions, extending toward space–air–ground–sea QKD networks through inventory-aware routing, dynamic link scheduling, and resource allocation.

Federated Intelligence and Cooperative Maritime Sensing 2025 – Present
*Collaboration with Prof. Xiaojuan Zhang, A*STAR Singapore* *Remote / Singapore*

- Investigated cooperative sensing and distributed decision-making in heterogeneous maritime UAV–USV networks, focusing on coupled communication, energy, queueing, and resource constraints under dynamic conditions.
- Developed federated reinforcement learning for cooperative resource control, integrating PPO-based policy learning with UAV-assisted aggregation under limited communication, device heterogeneity, and changing network conditions.
- Explored robust hierarchical federated fusion for maritime perception, integrating update-quality assessment, robust aggregation, and soft synchronization across heterogeneous edge devices with unreliable communication.

Honors & Awards

- **Best Paper Award**, 2025 4th International Conference on Artificial Intelligence, Internet of Things and Cloud Computing Technology (AIoTC), 2025.
- **National Encouragement Scholarship**, 2021 – 2023.
- Second Prize, The 10th "Datang Cup" Mobile Communication 5G Technology Competition in Zhejiang, 2023.
- Second Prize, Zhejiang Undergraduate Physics Experiment and Technology Innovation Competition, 2022.

Technical Skills

Programming: Python, MATLAB, C/C++
Frameworks & Tools: PyTorch, NumPy, Linux/SSH, LaTeX
Methods: Wireless resource allocation; semantic/task-oriented communication; federated learning; reinforcement and online learning; causal inference; network optimization; flow matching
Languages: Mandarin (native), English